Xuan Huang

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Educational Background

Shanghai Jiao Tong University **Sept. 2026 – Jun. 2031 (expected)**
Ph.D. in Electronic and Information Engineering

University of Electronic Science and Technology of China **Sept. 2022 – Jun. 2026 (expected)**
B.Eng in Communication Engineering

- **Rank – 4/189 (top 2.1%)**; **Average Score – 94.2/100**; **GPA – 3.93/4**.

Research Experiences

Efficient Token Compression for Multimodal Large Language Models **Dec. 2025 – Present**
Undergraduate Thesis Project & Research Intern, Ant Group

- **Streaming Token Redundancy Modeling:** Analyzed temporal redundancy and semantic repetition in streaming video tokens; developed a redundancy identification framework tailored for streaming scenarios.
- **Token Compression Algorithm Design:** Designed a pre-LLM token pruning strategy that preserves causal reasoning, leveraging semantic segmentation and cross-frame feature similarity to retain key tokens while reducing input size.
- **Streaming Inference Framework Implementation:** Built an experimental pipeline based on pretrained VLMs, integrating token compression with KV-cache-based streaming inference; evaluated compression ratio, latency, and task performance.

Self-powered Multimodal Emotion Recognition System **Dec. 2023 – Aug. 2024**
National Undergraduate Innovation and Entrepreneurship Program (Excellent Project)

- **Multimodal Fusion Model:** Built a dual-modal model with modality-specific/invariant feature fusion for speech and wavelet-based EEG preprocessing for ViT input.
- **IoT System Development:** Developed full-stack system with ESP32-based EEG acquisition, Huawei Cloud for IoT data storage, Alibaba Cloud for backend model deployment, and WeChat Mini Program frontend.
- Published a **First-author** paper which is accepted by *ICDT 2025*.
- **National First Prize** in Huawei IoT Design Competition (Top 3.6%).

Publications

Voice Recognition System for Speech-to-Text and GPT Communication Powered by Organic Photovoltaic
X. Huang, Y. Zhang, X. Zhang, X. Zhang, and H. Tang.
Accepted *by ICDT 2025.* [**First author**].

Mechanical Power Modeling and Energy Efficiency Maximization for Movable Antenna Systems
X. Wei, W. Mei, **X. Huang**, Z. Chen, and B. Ning.
Accepted *by IEEE GLOBECOM 2025.* [**Third author**]. [**PDF**].

Energy-Efficient Movable Antennas: Mechanical Power Modeling and Performance Optimization
X. Wei, W. Mei, **X. Huang**, Z. Chen, and B. Ning.
Accepted *by IEEE TWC.* [**Third author**]. [**PDF**].

Honors and Awards

Outstanding Graduate of Sichuan Province (Top 4%)

2025 Tencent First-Class Scholarship (“Chengdian Friend Scholarship”)

2024 **National Scholarship (Top 1.5%)** for Undergraduates

2023 & 2024 & 2025 UESTC Outstanding Student Scholarship (Top 8%)

2024 National IoT Design Competition — *National First Prize*

2024 Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM) — *National Second Prize*

2024 National English Competition for College Students (NECCS) — *National Second Prize*

2024 Mathematical Contest in Modeling (MCM) — *Honorable Mention*

Skills and Services

Technical skills: Experienced in Python, C/C++ and Matlab. Available for JavaScript and Verilog.
Language: Chinese (native); English: CET-4: 664, CET-6: 625.